EXCELLENCE

CLASS 11 - 2026-27

Physics | Units 05

Units and Measurement

Dimensional Questions and Error Combination

By - Excellence Programme

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Topics to be covered

01 Dimensional questions continued from Units 04

02 Error in a sum and a difference

03 Error in a product, a quotient and a power

04 Percentage error in energy, density and g

05 Error in a sine, a lens and an exponential

Where we left Units 04

Last class we finished significant figures, accuracy, precision, and the mean error of a pendulum. The next board in that deck is slide 49: matching powers of M, L and T.

We finish those dimensional questions first. After that we learn how an error grows when two measured quantities are added, multiplied, divided, or raised to a power.

Write the formula first. The formula tells you whether to add absolute errors or relative errors.

Class notes will be saved after teaching. This deck is for the board.

Matching powers still works

Write the unknown as a product of powers. Put dimensions on both sides. Equate the powers of M, of L and of T separately. A pure number such as 2π or 6π never comes out of this method.

If two quantities in the list have the same dimensions, they cannot both be used as separate base quantities. If a quantity is an angle, it is a pure number and cannot build a force or a time by itself.

Same dimensions, or a pure angle, means the method stops.

Electric field and magnetic field

The relation between [E] and [B] is

1. $[E] = [B]\,[L]\,[T]$
2. $[E] = [B]\,[L]^{-1}\,[T]$
3. $[E] = [B]\,[L]\,[T]^{-1}$
4. $[E] = [B]\,[L]^{-1}\,[T]^{-1}$

Solution: Electric field and magnetic field

In an electromagnetic wave, E = c B. Speed c carries the extra dimensions.

Step 1: $[c] = L\,T^{-1}$, so $[E] = [B] \times L\,T^{-1}$.

Step 2: That is $[E] = [B]\,[L]\,[T]^{-1}$.

Final answer: option 3

Rough work / extra work

Permittivity and permeability

The relation between [ε_0] and [μ_0] is

1. $[\mu_0] = [\varepsilon_0] [L]^2 [T]^{-2}$
2. $[\mu_0] = [\varepsilon_0] [L]^{-2} [T]^2$
3. $[\mu_0] = [\varepsilon_0]^{-1} [L]^2 [T]^{-2}$
4. $[\mu_0] = [\varepsilon_0]^{-1} [L]^{-2} [T]^2$

Solution: Permittivity and permeability

Use $c = 1 / \sqrt{(\mu_0 \varepsilon_0)}$. Square both sides.

Step 1: $\mu_0 \varepsilon_0 = 1 / c^2$, so $[\mu_0]\,[\varepsilon_0] = L^{-2} T^2$.

Step 2: $[\mu_0] = [\varepsilon_0]^{-1} [L]^{-2} [T]^2$.

Final answer: option 4

Same dimensions as S

S = (E × B) / μ_0. Its dimensions match which of these?

1. Energy / (charge × current)
2. Force / (length × time)
3. Energy / volume
4. Power / area

Solution: Same dimensions as S

Build [S] from the known formulas, then compare.

Step 1: $[E] = M\,L\,T^{-3}\,I^{-1}$, $[B] = M\,T^{-2}\,I^{-1}$, $[\mu_0] = M\,L\,T^{-2}\,I^{-2}$.

Step 2: $[E\,B / \mu_0] = M\,T^{-3}$.

Step 3: Force / (length × time) = $M\,T^{-3}$. Power / area = $M\,T^{-3}$.

Final answer: options 2 and 4

Rough work / extra work

Mass and angular momentum are pure numbers

In a new system, mass and angular momentum are dimensionless, and length has dimension L. Which statements are correct?

1. Dimension of force is L^{-3}
2. Dimension of energy is L^{-2}
3. Dimension of power is L^{-5}
4. Dimension of linear momentum is L^{-1}

Solution: Mass and angular momentum are pure numbers

Angular momentum m v r is a pure number, and m is a pure number, so velocity must cancel the length.

Step 1: [v] = L^{-1}. Then [time] = length / speed = L^{2}.

Step 2: [force] = [mass × acceleration] = L^{-3}. [energy] = force × length = L^{-2}.

Step 3: [power] = energy / time = L^{-4}, not L^{-5}. [momentum] = m v = L^{-1}.

Correct: options 1, 2 and 4. Option 3 is wrong.

Young's modulus in F, A and V

Speed V, area A and force F are the base quantities. [Young's modulus] is

1. $F A^{-1} V^{0}$
2. $F A^{2} V^{-1}$
3. $F A^{2} V^{-2}$
4. $F A^{2} V^{-3}$

Solution: Young’s modulus in F, A and V

Young’s modulus is stress / strain, and strain is a pure number.

Step 1: Y = F / A.

Step 2: Speed is not needed, so its power is 0.

Final answer: $F\ A^{-1}\ V^{0}$ (option 1)

Density from force, velocity and time

F, V and T are base quantities. The dimensional formula of density is

1. $F\ V^{4}\ T^{-6}$
2. $F\ V^{-4}\ T^{-2}$
3. $F^{2}\ V^{-2}\ T^{6}$
4. $F\ V^{-2}\ T^{2}$

Solution: Density from force, velocity and time

Assume density = $F^a\ V^b\ T^c$ and match M, L, T.

Step 1: Mass: a = 1. Length: a + b = -3, so b = -4.

Step 2: Time: -2a - b + c = 0 → -2 - (-4) + c = 0 → c = -2.

Final answer: $F\ V^{-4}\ T^{-2}$ (option 2)

Rough work / extra work

Viscosity from momentum, area and time

P, A and T are base quantities. The dimensional formula of viscosity is

1. $[P\ A^{-1}]$
2. $[P\ A\ T^{-1}]$
3. $[P\ A^{-1}\ T]$
4. $[P\ A^{-1}\ T^{-1}]$

Solution: Viscosity from momentum, area and time

From $F = 6\pi \eta r v$, the dimensions of η are $M L^{-1} T^{-1}$.

Step 1: $\eta = P^a A^b T^c$. Mass gives a = 1.

Step 2: Length: $1 + 2b = -1$, so $b = -1$. Time: $-1 + c = -1$, so $c = 0$.

Final answer: $[P A^{-1}]$ (option 1)

Mass from time, velocity and angular momentum

Express mass using time t, velocity v and angular momentum J.

Solution: Mass from time, velocity and angular momentum

$[J] = M\,L^2\,T^{-1}$. Assume $M = J^a\,v^b\,t^c$.

Step 1: Mass: a = 1. Length: 2 + b = 0, so b = -2.

Step 2: Time: -1 - (-2) + c = 0, so c = -1.

Mass = $J / (v^2\,t)$

Rough work / extra work

Young's modulus from c, h and G

Young's modulus is a product of powers of c, h and G. Which powers are correct?

1. $\alpha = 7, \beta = -1, \gamma = -2$
2. $\alpha = -7, \beta = -1, \gamma = -2$
3. $\alpha = 7, \beta = -1, \gamma = 2$
4. $\alpha = -7, \beta = 1, \gamma = -2$

Solution: equations for Y

$[Y] = M\,L^{-1}\,T^{-2}$, $[c] = L\,T^{-1}$, $[h] = M\,L^{2}\,T^{-1}$, $[G] = M^{-1}\,L^{3}\,T^{-2}$.

Step 1: Mass: $\beta - \gamma = 1$.

Step 2: Length: $\alpha + 2\beta + 3\gamma = -1$.

Step 3: Time: $-\alpha - \beta - 2\gamma = -2$, so $\alpha + \beta + 2\gamma = 2$.

Solution: the three powers

From mass, $\beta = 1 + \gamma$. Put that into the time equation.

Step 1: $\alpha + (1 + \gamma) + 2\gamma = 2 \rightarrow \alpha + 3\gamma = 1$.

Step 2: Length then gives $\alpha + 5\gamma = -3$. Subtract: $2\gamma = -4$, so $\gamma = -2$.

Step 3: $\alpha = 1 - 3(-2) = 7$. $\beta = 1 + (-2) = -1$.

$\alpha = 7, \beta = -1, \gamma = -2$ (option 1)

A leaking balloon

Gas speed depends on the radius through a power a. The emptying time T depends on surface tension S, outlet area A, density ρ and starting radius R through powers α, β, γ and δ.

1. $a = 1/2, \alpha = 1/2, \beta = -1, \gamma = 1, \delta = 3/2$
2. $a = -1/2, \alpha = -1/2, \beta = -1, \gamma = -1/2, \delta = 5/2$
3. $a = -1/2, \alpha = -1/2, \beta = -1, \gamma = 1/2, \delta = 7/2$
4. $a = 1/2, \alpha = 1/2, \beta = -1/2, \gamma = 1/2, \delta = 7/2$

Solution: speed of the gas

The speed of the leaving gas is fixed by S, ρ and the present radius r.

Step 1: $v \propto S^{1/2} \rho^{-1/2} r^{-1/2}$.

Step 2: Mass cancels. What remains is a speed, and the power of r is -1/2. So a = -1/2.

Rough work / extra work

Solution: time to empty

Volume leaves through area A, so the radius falls as $dr/dt \propto A\, v / r^2$.

Step 1: With $v \propto r^{-1/2}$, we get $dr/dt \propto A\, S^{1/2}\, \rho^{-1/2}\, r^{-5/2}$.

Step 2: Time to fall from R to 0 is $T \propto R^{7/2} / (A\, S^{1/2}\, \rho^{-1/2})$.

Step 3: So $\alpha = -1/2$, $\beta = -1$, $\gamma = 1/2$, $\delta = 7/2$.

Final answer: option 3

Three cases where dimensions refuse

- Torque, energy and work have the same dimensions. Using all three as base quantities cannot produce a new formula for time.

- If g is said to depend on force, energy, mass and velocity, that is four quantities and only three equations (M, L, T). The powers are not unique.

- If force is said to depend on an angle, a velocity and an energy, the angle is a pure number. It cannot supply the dimensions of force.

Write “not possible” and name the reason. That is a full answer.

Which equation cannot be fully derived

Which equation cannot be obtained from dimensions alone?

1. $F = 6\pi \eta r v$
2. $\theta = \omega t$
3. $P = \rho g h$
4. $v = d / t$

Solution: Which equation cannot be fully derived

Dimensions can give the powers. They cannot give a pure number such as 6π.

Step 1: F, η, r and v can be matched, but 6π stays unknown.

Step 2: θ = ω t, P = ρ g h and v = d / t are products of powers. Their number in front is 1.

Final answer: F = 6π η r v (option 1)

Which error to write first

An absolute error has the same unit as the quantity. A relative error is the absolute error divided by the measured value, so it has no unit. Percentage error is the relative error × 100.

For a sum or a difference, write the absolute error directly. For a product, a quotient, or a power, write the relative error directly, then multiply by the answer if the absolute error is asked.

Report the result as (value ± absolute error) with the unit.

Rough work / extra work

Sum and difference

If Y = A + B, or Y = A − B, the maximum absolute error is the sum of the two absolute errors. The errors are added even when the quantities are subtracted.

ΔY = ΔA + ΔB

Relative error of a sum = (ΔA + ΔB) / (A + B)

Relative error of a difference = (ΔA + ΔB) / (A − B)

Two close numbers subtracted leave a small answer and a large percentage error.

Product, quotient and power

If Y = A B, or Y = A / B, the relative errors are added. The same rule covers a power: each relative error is multiplied by the size of its power.

$$\Delta Y / Y = \Delta A / A + \Delta B / B$$

$$Y = k A^m B^n \quad \rightarrow \quad \Delta Y / Y = |m| \, \Delta A/A + |n| \, \Delta B/B$$

A fixed number such as 1/2, 2 or π has no measurement error. Its power still multiplies the relative error of the measured quantity.

From the derivative

A small change follows the derivative. If $y = x^n$, then $\Delta y / y = |n| \, \Delta x / x$. That is the power rule written in one line.

$y = \sin\theta \quad \rightarrow \quad \Delta y = |\cos\theta| \, \Delta\theta$, with $\Delta\theta$ in radians

$y = 1/x \quad \rightarrow \quad \Delta y / y = \Delta x / x$

Take every power as a positive number when you want the maximum error.

Percentage error in X

$X = M^a L^b T^{-c}$. Percentage errors in M, L and T are α, β and γ, and they are small. The maximum percentage error in X is

1. $(a\alpha - b\beta + c\gamma)$ %
2. $(a\alpha + b\beta + c\gamma)$ %
3. $(a\alpha + b\beta - c\gamma)$ %
4. $(\alpha + \beta + \gamma)$ %

Solution: Percentage error in X

The power of T is −c. For the maximum error we use the size of that power, which is c.

Step 1: $\Delta X / X \times 100 = |a|\alpha + |b|\beta + |-c|\gamma$.

Step 2: With positive a, b and c this is $(a\alpha + b\beta + c\gamma)$ %.

Final answer: option 2

Rough work / extra work

A product of powers

$P = a^2 b^3 / (c \sqrt{d})$. Percentage errors in a, b, c and d are 1%, 2%, 3% and 4%. The percentage error in P is

1. 13 %
2. 16 %
3. 12 %
4. 10 %

Solution: A product of powers

√d means d raised to 1/2. Each percentage is multiplied by the size of its power.

Step 1: 2×1 + 3×2 + 1×3 + (1/2)×4

Step 2: = 2 + 6 + 3 + 2 = 13.

Final answer: 13 % (option 1)

Area of a rectangular plate

A plate is (2.00 ± 0.02) cm by (1.00 ± 0.01) cm. The percentage error in its area is

1. 1 %
2. 2 %
3. 3 %
4. 4 %

Solution: Area of a rectangular plate

Area = length × breadth, so the relative errors are added.

Step 1: $\Delta A / A = 0.02/2.00 + 0.01/1.00 = 0.01 + 0.01$.

Step 2: Percentage error = $0.02 \times 100 = 2$ %.

Final answer: 2 % (option 2)

Kinetic energy

m = (5.0 ± 0.5) kg and v = (20.0 ± 0.4) m/s. The kinetic energy with its error is

1. (1000 ± 140) J
2. (1000 ± 0.14) J
3. (500 ± 140) J
4. (1000 ± 70) J

Solution: Kinetic energy

$K = (1/2)\, m\, v^2$. The 1/2 is exact, so it adds no error term. The power of v is 2.

Step 1: $K = (1/2)\times 5\times 400 = 1000$ J.

Step 2: $\Delta K / K = 0.5/5 + 2\times(0.4/20) = 0.10 + 0.04 = 0.14$.

Step 3: $\Delta K = 0.14 \times 1000 = 140$ J.

Final answer: (1000 ± 140) J (option 1)

Rough work / extra work

Side of a square park

The area of a square park is (100 ± 0.2) m^2. The side of the park is

1. (10.00 ± 0.01) m
2. (10.00 ± 0.02) m
3. (10.0 ± 0.1) m
4. (10.00 ± 0.20) m

Solution: Side of a square park

Find the side first. Area = side^2, so the relative error in the side is half the relative error in the area.

Step 1: Side = √100 = 10 m.

Step 2: $\Delta l / l = (1/2) \times (0.2 / 100) = 0.001$.

Step 3: $\Delta l = 0.001 \times 10 = 0.01$ m.

Final answer: (10.00 ± 0.01) m (option 1)

Pressure

$P = F / L^2$. The percentage error in F is 4 % and the percentage error in L is 2 %. The percentage error in P is

1. 2 %
2. 4 %
3. 6 %
4. 8 %

Solution: Pressure

L is squared, so its percentage error is doubled and then added to the percentage error in F.

Step 1: $\Delta P / P \times 100 = 4 + 2 \times 2 = 8$.

Final answer: 8 % (option 4)

Rough work / extra work

Density of a wire

$\rho = m / (\pi r^2 L)$. $m = (0.4 \pm 0.01)$ g, $L = (8 \pm 0.04)$ cm, $r = (6 \pm 0.03)$ mm. The percentage error in density is

1. 1 %
2. 2 %
3. 4 %
4. 5 %

Solution: Density of a wire

π is exact. Each relative error uses its own unit, so millimetres and centimetres can stay as they are.

Step 1: $(0.01/0.4)\times100 + 2\times(0.03/6)\times100 + (0.04/8)\times100$

Step 2: $= 2.5 + 1.0 + 0.5 = 4.$

Final answer: 4 % (option 3)

Relative error in Z

$Z = A^2 B^3 / C^4$. The maximum relative error in Z is

1. $\Delta A/A + \Delta B/B + \Delta C/C$
2. $2\Delta A/A + 3\Delta B/B - 4\Delta C/C$
3. $2\Delta A/A + 3\Delta B/B + 4\Delta C/C$
4. $2\Delta A/A + 3\Delta B/B + \Delta C/C$

Solution: Relative error in Z

The power of C is −4. For the maximum error that power is taken as 4, and the term is added.

Step 1: $\Delta Z / Z = 2\,\Delta A/A + 3\,\Delta B/B + 4\,\Delta C/C$.

Final answer: option 3

Torque

Torque has dimensions $M L^2 T^{-2}$. If M, L and T are each measured to 5 % accuracy, the percentage error in torque is

1. 5 %
2. 25 %
3. 15 %
4. 10 %

Solution: Torque

The powers are 1 for mass, 2 for length and 2 for time.

Step 1: $1\times5 + 2\times5 + 2\times5 = 5 + 10 + 10 = 25$.

Final answer: 25 % (option 2)

Rough work / extra work

g from a pendulum

In $T = 2\pi \sqrt{(l/g)}$, the relative errors in l and T are α and β. The percentage error in g is

1. $(\alpha + \beta) \times 100$
2. $(\alpha + \beta/2) \times 100$
3. $(\alpha/2 + \beta) \times 100$
4. $(\alpha + 2\beta) \times 100$

Solution: g from a pendulum

Square the period formula and solve for g. The $4\pi^2$ is exact.

Step 1: $g = 4\pi^2 l / T^2$.

Step 2: $\Delta g / g = \Delta l/l + 2\,\Delta T/T = \alpha + 2\beta$.

Step 3: Percentage error = $(\alpha + 2\beta) \times 100$.

Final answer: option 4

A measured pendulum

l = 10 cm, known to 1 mm. The time for 200 oscillations is 100 s, read on a clock of 1 s resolution. The percentage error in g is

1. 3 %
2. 5 %
3. 4 %
4. 2 %

Solution: A measured pendulum

The number of oscillations is exact, so the relative error in the period equals the relative error in the total time.

Step 1: $\Delta l / l = 0.1\ \text{cm} / 10\ \text{cm} = 0.01$.

Step 2: $\Delta T / T = 1\ \text{s} / 100\ \text{s} = 0.01$.

Step 3: $\Delta g / g = 0.01 + 2 \times 0.01 = 0.03$, so the percentage error is 3 %.

Final answer: 3 % (option 1). The fractional error is 0.03.

Rough work / extra work

Error in sin θ

$y = \sin\theta$. At $\theta = \pi/6$ the percentage error in θ is 2 %. Find the percentage error in y.

Solution: Error in sin θ

$\Delta y = |\cos\theta|\,\Delta\theta$, and $\Delta\theta$ must be in radians. A 2 % error means $\Delta\theta = 0.02 \times \theta$.

Step 1: $\Delta y / y = |\cot\theta| \times \Delta\theta = \sqrt{3} \times 0.02 \times (\pi/6)$.

Step 2: Percentage error $= \sqrt{3} \times 0.02 \times (\pi/6) \times 100 = \pi/\sqrt{3}$ %.

Final answer: $\pi/\sqrt{3}$ % (about 1.81 %)

Where the percentage error is smallest

$y = m \tan \theta$, and m is exact. The absolute error in θ is the same at every angle. The percentage error in y is smallest at

1. $\theta = 30°$
2. $\theta = 45°$
3. $\theta = 60°$
4. $\theta = 90°$

Solution: Where the percentage error is smallest

Differentiate $y = m \tan \theta$. The constant m cancels in the relative error.

Step 1: $\Delta y / y = 2\,\Delta\theta / \sin 2\theta$.

Step 2: With $\Delta\theta$ fixed, this is smallest when $\sin 2\theta = 1$.

Step 3: $2\theta = 90°$, so $\theta = 45°$.

Final answer: $\theta = 45°$ (option 2)

Series, product and difference

$R_1 = (10 \pm 1)\ \Omega$ and $R_2 = (20 \pm 1)\ \Omega$.

Series: $R = 30\ \Omega$, $\Delta R = 2\ \Omega$, percentage $= (2/30)\times 100 = 6.67\ \%$.

Product: $R_1 R_2 = 200$, relative error $= 0.10 + 0.05 = 0.15$.

Absolute error in the product $= 200 \times 0.15 = 30$.

Difference: $R = 10\ \Omega$, $\Delta R = 2\ \Omega$, percentage $= 20\ \%$.

Quotient $R_2 / R_1 = 2$ also has relative error 0.15, so its absolute error is $0.30\ \Omega$.

Rough work / extra work

Parallel resistors and a thin lens

For two resistors in parallel, 1/R = 1/R_1 + 1/R_2. Differentiating gives the absolute error in R from the squares of the two resistances.

$$\Delta R = R^2 \left(\Delta R_1 / R_1^2 + \Delta R_2 / R_2^2 \right)$$

A thin lens with a real image uses the same shape: $1/f = 1/u + 1/v$, so $\Delta f = f^2 (\Delta u/u^2 + \Delta v/v^2)$.

Series capacitors, and 1/f = 1/u + 1/v, follow this same squared form.

Two resistors in parallel

$R_1 = (20 \pm 2)\ \Omega$ and $R_2 = (5 \pm 1)\ \Omega$ are in parallel. Find the percentage error and the absolute error in the equivalent resistance.

Solution: Two resistors in parallel

First find the equivalent resistance, then use the parallel error formula.

Step 1: $R = (20 \times 5)/(20+5) = 4\ \Omega$.

Step 2: $\Delta R / R = 4\,(2/20^2 + 1/5^2) = 4\,(0.005 + 0.04) = 0.18$.

Step 3: Percentage error = 18 %. Absolute error = $0.18 \times 4 = 0.72\ \Omega$.

Final answer: 18 % and 0.72 Ω

Focal length of a convex lens

Object distance u = (10 ± 0.1) cm and image distance v = (20 ± 0.2) cm. The error in the focal length is n %. Find n.

Solution: Focal length of a convex lens

Use $1/f = 1/u + 1/v$ with both distances taken positive.

Step 1: $1/f = 1/10 + 1/20 = 3/20$, so $f = 20/3$ cm.

Step 2: $\Delta f / f = f\,(0.1/10^2 + 0.2/20^2) = (20/3)(0.001 + 0.0005)$.

Step 3: $(20/3)\times 0.0015 = 0.01$, so the percentage error is 1 %.

Final answer: n = 1

Rough work / extra work

Error in d from 2d sin θ = λ

λ is known exactly, and the error in θ is the same at every angle. As θ increases from 0° to 90°, which statement is true?

1. The absolute error in d stays constant
2. The absolute error in d increases
3. The fractional error in d stays constant
4. The fractional error in d decreases

Solution: Error in d from 2d sin θ = λ

d = λ / (2 sin θ). Differentiating with respect to θ gives the fractional error.

Step 1: $\Delta d / d = |\cot \theta| \Delta\theta$.

Step 2: As θ rises from 0° toward 90°, cot θ falls, so the fractional error falls.

Final answer: option 4

Error in the ratio r

r = (1 − a)/(1 + a), and the error in the dimensionless quantity a is Δa. Then Δr equals

1. $\Delta a / (1 + a)^2$
2. $-2\,\Delta a / (1 + a)^2$
3. $2\,\Delta a / (1 - a)^2$
4. $2\,a\,\Delta a / (1 - a^2)$

Solution: Error in the ratio r

Differentiate r with respect to a. The two minus signs in the quotient rule add.

Step 1: $dr/da = -2 / (1 + a)^2$.

Step 2: $\Delta r = -2\,\Delta a / (1 + a)^2$.

Final answer: option 2. The size of the error is $2\,\Delta a / (1 + a)^2$.

Energy that falls with time

$E = A^2 \exp(-\alpha t)$, with $\alpha = 0.2\ s^{-1}$. The error in A is 1.25 % and the error in t is 1.50 %. At t = 5 s, the percentage error in E is

Solution: Energy that falls with time

Take the logarithm. The power of A contributes 2. The exponential contributes α × the absolute error in t.

Step 1: $\Delta t = 0.015 \times 5 = 0.075$ s, so $\alpha \Delta t = 0.2 \times 0.075 = 0.015$.

Step 2: Percentage from A: $2 \times 1.25 = 2.5$. From time: 1.5.

Step 3: Total percentage error = 2.5 + 1.5 = 4.

Final answer: 4 %

Rough work / extra work

Current in a diode

I = (exp(1000 V/T) – 1) mA. At 300 K the current is 5 mA, and the voltage error is 0.01 V. The error in the current is

1. 0.2 mA
2. 0.02 mA
3. 0.5 mA
4. 0.05 mA

Solution: Current in a diode

The measured current fixes the exponential. Then differentiate with respect to V.

Step 1: $\exp(1000\ V/T) = I + 1 = 6$.

Step 2: $\Delta I = 6 \times (1000/300) \times 0.01 = 6 \times (10/3) \times 0.01$.

Step 3: $\Delta I = 0.2$ mA.

Final answer: 0.2 mA (option 1)

g from a periodic motion

$T = 2\pi \sqrt{[7(R - r)/(5g)]}$. $R = (60 \pm 1)$ mm, $r = (10 \pm 1)$ mm. Five periods are 0.52, 0.56, 0.57, 0.54 and 0.59 s. The watch reads to 0.01 s. Which statements are true?

1. The error in r is 10 %
2. The error in T is 3.57 %
3. The error in T is 2 %
4. The error in g is 11 %

Solution: the mean period

Mean of the five times: (0.52 + 0.56 + 0.57 + 0.54 + 0.59)/5 = 0.556 s, taken as 0.56 s.

Step 1: Mean absolute error ≈ 0.02 s, larger than the least count 0.01 s.

Step 2: $\Delta T / T \times 100 = (0.02 / 0.56) \times 100 = 3.57$ %.

Step 3: Error in r: (1/10)×100 = 10 %. The 2 % statement is false.

Statements 1 and 2 are true.

Rough work / extra work

Solution: the error in g

From the period formula, g depends on $(R - r)$ and on T^2.

Step 1: $R - r = 50$ mm, $\Delta(R - r) = 1 + 1 = 2$ mm, so 4 %.

Step 2: The T^2 term contributes $2 \times 3.57\ \% = 7.14\ \%$.

Step 3: Percentage error in g = $4 + 7.14 \approx 11\ \%$.

True statements: 1, 2 and 4.

A reading written as 2.5 g

When a mass is written 2.5 g and no separate instrument error is given, the last written digit is the uncertain one. The absolute error is one unit in that place.

Absolute error = 0.1 g.

Percentage error = (0.1 / 2.5) × 100 = 4 %.

If the instrument’s least count is given, use that least count as the absolute error.

Units 05 lock lines

Sum or difference: add the absolute errors.

Product or quotient: add the relative errors.

A power multiplies its own relative error. Use the positive size of the power.

Parallel resistors and a lens use $\Delta R = R^2 (\Delta R_1/R_1^2 + \Delta R_2/R_2^2)$.

Dimensions still cannot produce a pure number such as 6π. That limit from the first half of class stays.

Lines worth writing in full

Maximum absolute error in a sum or a difference is $\Delta A + \Delta B$.

Maximum relative error in a product or a quotient is $\Delta A/A + \Delta B/B$.

For $Y = A^m B^n$, the maximum relative error is $|m|\Delta A/A + |n|\Delta B/B$.

A result is written (measured value ± absolute error) unit.

These four lines cover the error chapter in the board paper.

Before the next class

1. A = (2.0 ± 0.1) and B = (3.0 ± 0.2). Find A B with its absolute error.

2. A sphere's mass has 2 % error and its diameter has 1 % error. Find the percentage error in density.

3. R_1 = (100 ± 3) Ω and R_2 = (200 ± 4) Ω in series. Find the percentage error.

4. For a pendulum, l has 1 % error and T has 2 % error. Find the percentage error in g.

Next class: metre scale, vernier callipers and screw gauge.

Rough work / extra work